

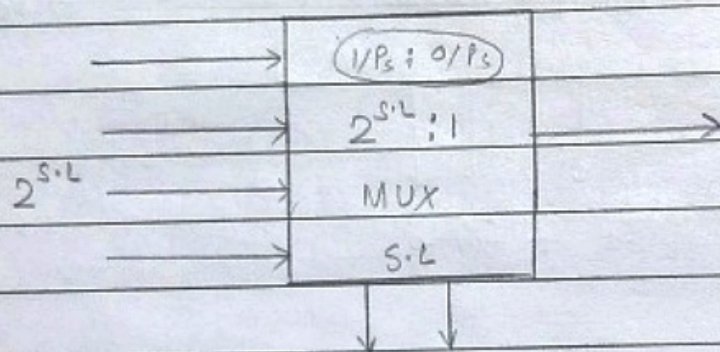
* Multiplexer (MUX)

It is a data selector which selects only 1 input depending upon the value of selected lines. It has many inputs but only 1 output. That's why it's also called many to one combinational circuit.

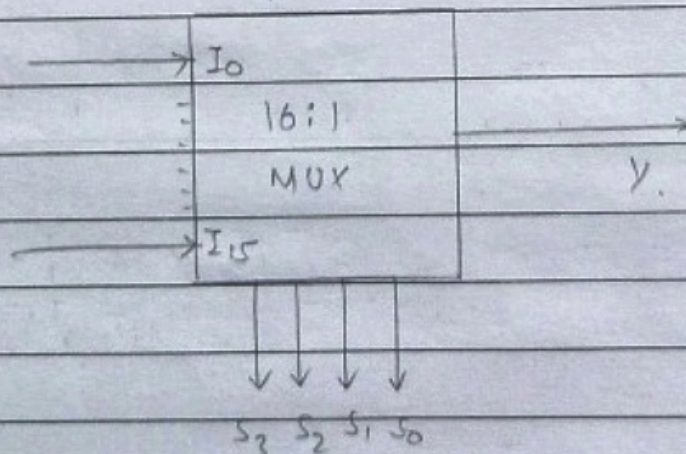
$$\text{Inputs} = m = 2^{S.L}$$

(selected lines)

Its value depends on no. of inputs.

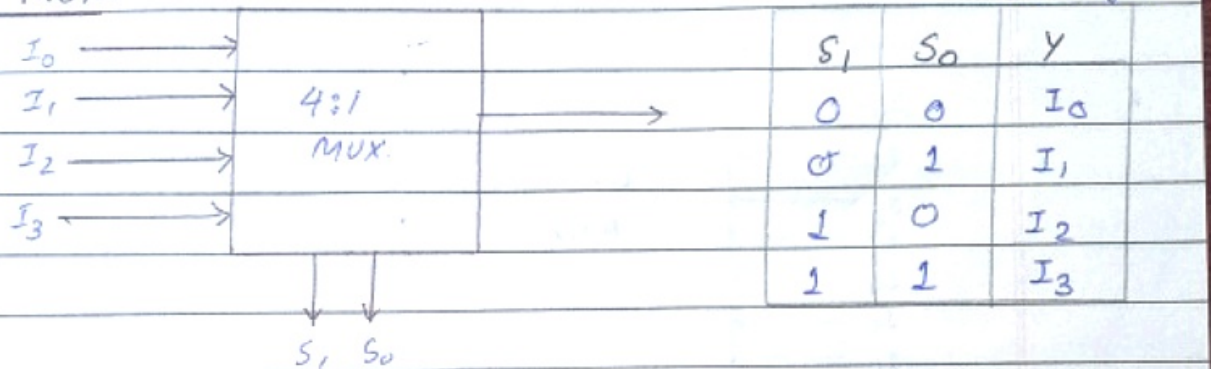


Eg:



decimal equivalent
of
(binary)

* 4:1 MUX.

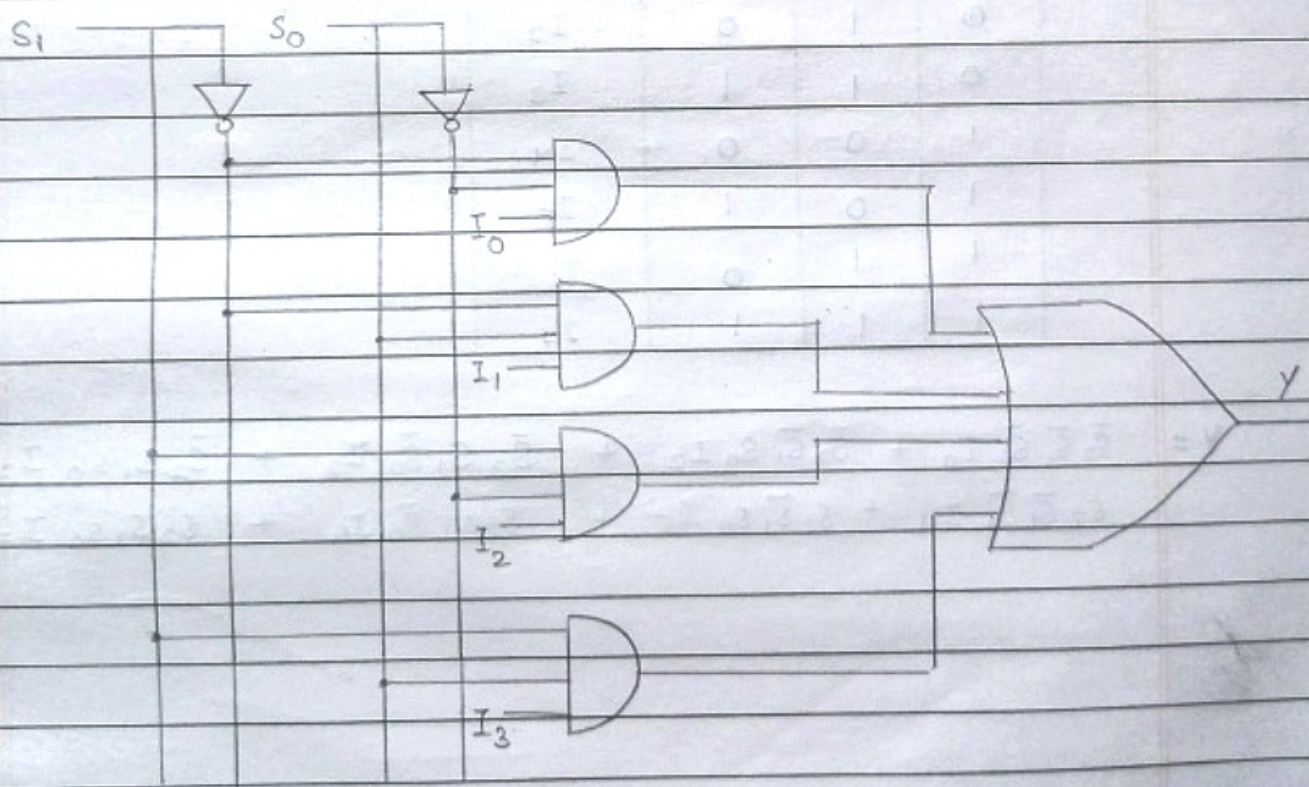


$$Y = \bar{S}_1 \bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 +$$

$$S_1 \bar{S}_0 I_2 + S_1 S_0 I_3$$

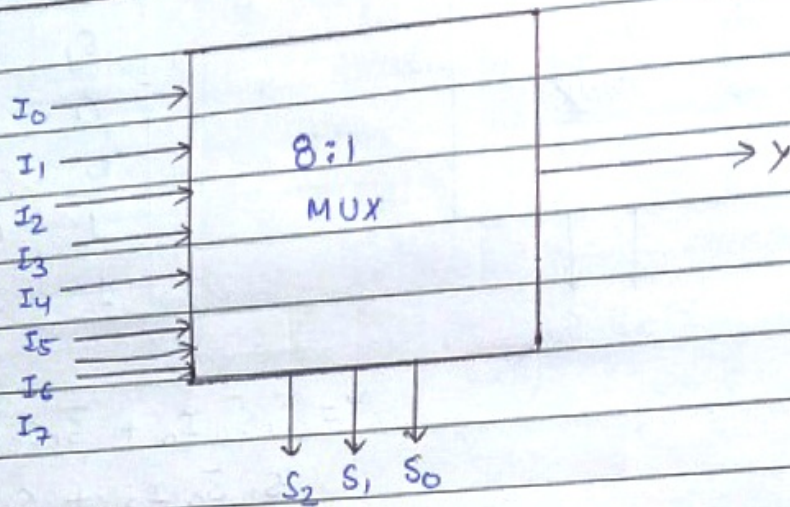
(minterms $\rightarrow$ SOP)

$$Y = 0 \times 1 + 0 \times 0 + 1 \times \bar{0} I_2 + 1 \times 0 I_3 \quad \text{when } S_1 = 1, S_0 = 0$$



In multiplexer, we get O/P in SOP form (minterm)

* 8:1 MUX

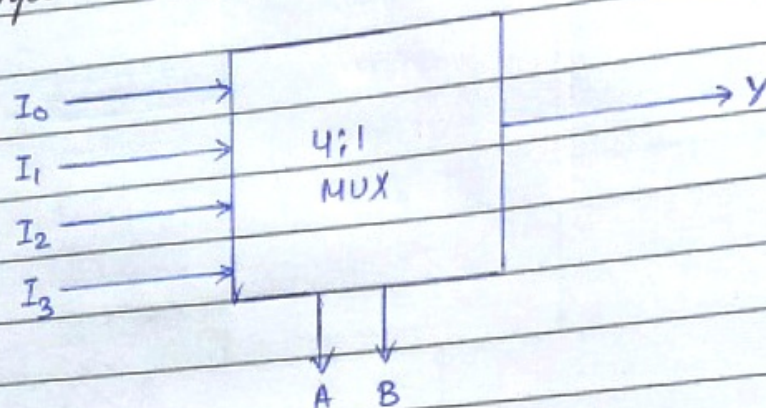


S_2	S_1	S_0	Y
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

$$Y = \bar{S}_2 \bar{S}_1 \bar{S}_0 I_0 + \bar{S}_2 \bar{S}_1 S_0 I_1 + \bar{S}_2 S_1 \bar{S}_0 I_2 + \bar{S}_2 S_1 S_0 I_3 + S_2 \bar{S}_1 \bar{S}_0 I_4 + S_2 \bar{S}_1 S_0 I_5 + S_2 S_1 \bar{S}_0 I_6 + S_2 S_1 S_0 I_7$$

Ques.

Implement $Y(A,B) = \sum m(0,1,3)$ using 4:1 MUX.



A	B	$Y (\sum m(0,1,3))$
0	0	1
0	1	1
1	0	0
1	1	1

For 4:1 MUX

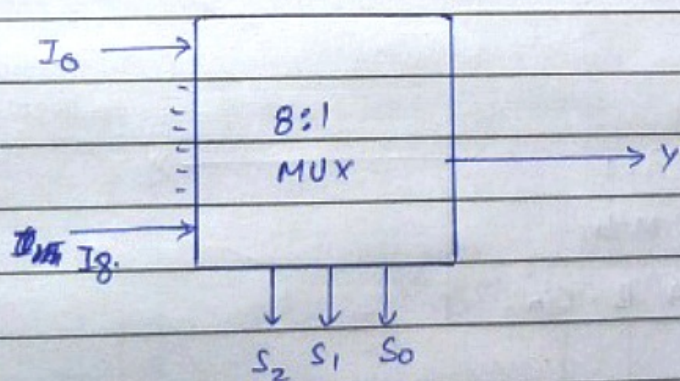
$$Y = \bar{S}_1 \bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1 \bar{S}_0 I_2 + S_1 S_0 I_3$$

Given, $I_0 = 1, I_1 = 1, I_2 = 0, I_3 = 1$

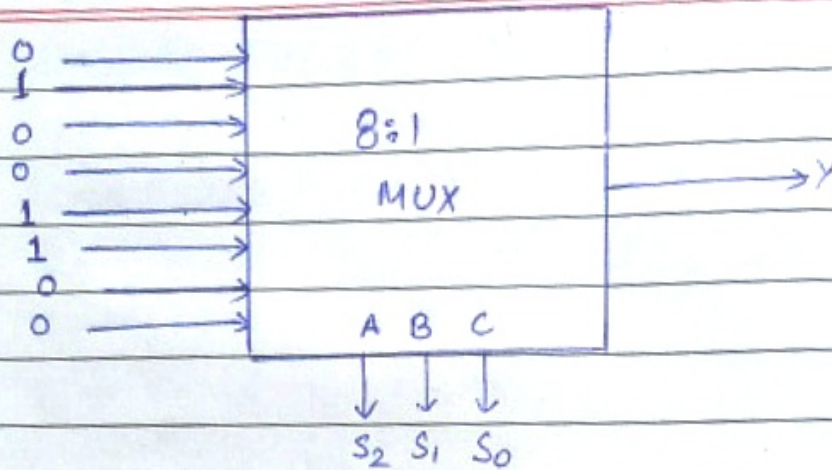
$$Y = \bar{S}_1 \bar{S}_0 + \bar{S}_1 S_0 + 0 + S_1 S_0$$

#NOTE: To implement any boolean funcⁿ using MUX, connect all the minterms which are given in problem statement / boolean funcⁿ with +5 V (1) otherwise with Ground 0V (0).

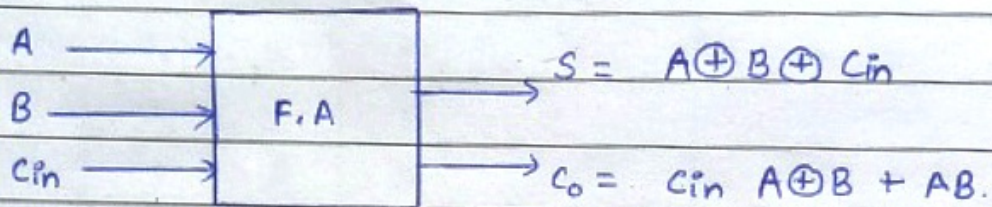
Ques. Implement $Y(A, B, C) = \pi M(0, 1, 3, 6, 7)$ using 8:1 MUX.
 $Y(A, B, C) = \Sigma m(2, 4, 5, 8, 9, 10, 11, 12, 13, 14, 15)$



$\Sigma m(1,4,5)$

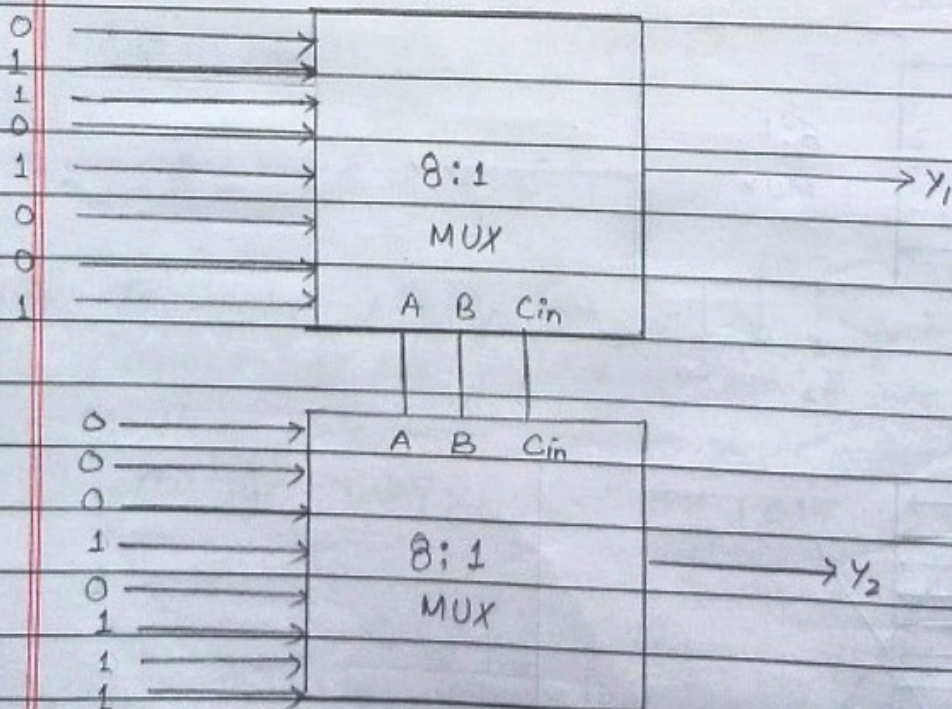


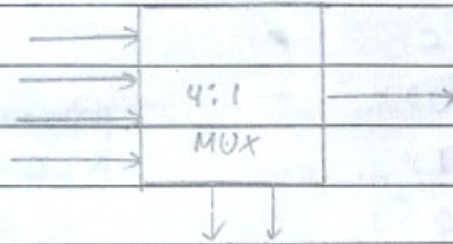
Ques. Implement full adder using 8:1 MUX.



$$C_o = \Sigma m(3, 5, 6, 7)$$

$$S = \Sigma m(1, 2, 4, 7)$$



$Y(A, B, C)$ Ques- Implement $Y = \sum m(1, 2, 5, 6)$ using 4:1 MUXonly 2 S.L ($(A, B)/(B, C)/(C, A)$)

Step 1: Truth table

 $\textcircled{Y}$

A

B

C

Y

 $I_0 = C$

0

0

0

0

0

0

1

1

 $I_1 = \bar{C}$

0

1

0

1

0

1

1

0

 $I_2 = C$

1

0

0

0

1

0

1

1

 $I_3 = \bar{C}$

1

1

0

1

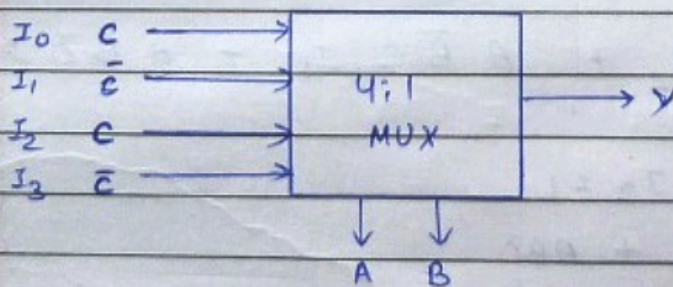
1

1

1

0

Case I: When A, B are selected lines



$\Sigma m(1, 2, 5, 6)$

Case II: when B, C are selected lines,

bits	4	2	1	
Y	A	B	C	
$I_0 = 0$	0	0	0	I_0
$I_1 = 1$	1	0	0	I_1
$I_2 = 1$	1	0	1	I_2
$I_3 = 0$	0	0	1	I_3
	0	1	0	I_0
	1	1	0	I_1
	1	1	1	I_2
	0	1	1	I_3

whatever value of A,
 $Y = 0$ when I_0 is
 selected, I_3

$Y = 1$ when I_1, I_2
 is selected.

we need to find relation
 between I and A

A \ BC	00	01	11	10
0		1		1
1		1		1

$$Y = \bar{B}C + B\bar{C}$$

(no affect of A on Y)

(cross-checked using
 K-map)

from truth table,

(minterms)

$$Y = \bar{A}\bar{B}C I_1 + \bar{A}B\bar{C} I_2 + A\bar{B}C I_1 + AB\bar{C} I_2$$

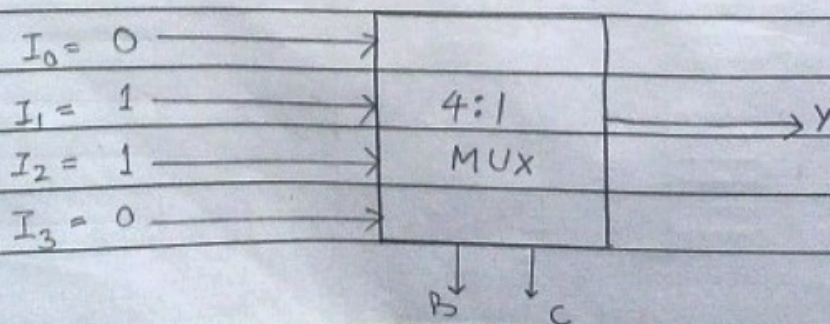
we found,

$$I_1 = 1, I_2 = 1$$

$$= \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}C + AB\bar{C}$$

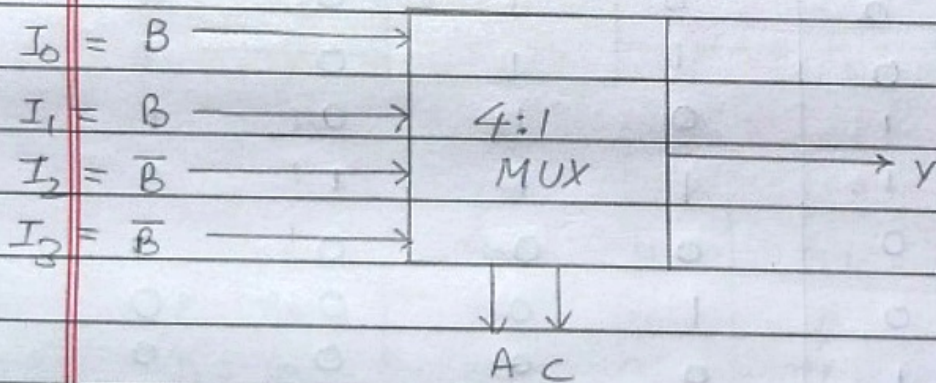
$$= \bar{B}C(\bar{A} + A) + B\bar{C}(\bar{A} + A)$$

$$Y = \bar{B}C + B\bar{C}$$



Case II When A, C are S.L.

		C	A	B	Y
Y=B	I ₀	0	0	0	0
	I ₂	1	0	0	1
Y=B̄	I ₀	0	0	1	1
	I ₂	1	0	1	0
Y=B	I ₁	0	1	0	0
	I ₃	1	1	0	1
Y=B̄	I ₁	0	1	1	1
	I ₃	1	1	1	0



Q1. Implement $Y = \bar{A} + BC$ using

a) 8:1 MUX

b) 4:1 MUX

Q2. Implement full subtractor using 4:1 MUX. when A, B are selected lines. (in terms of Bin)

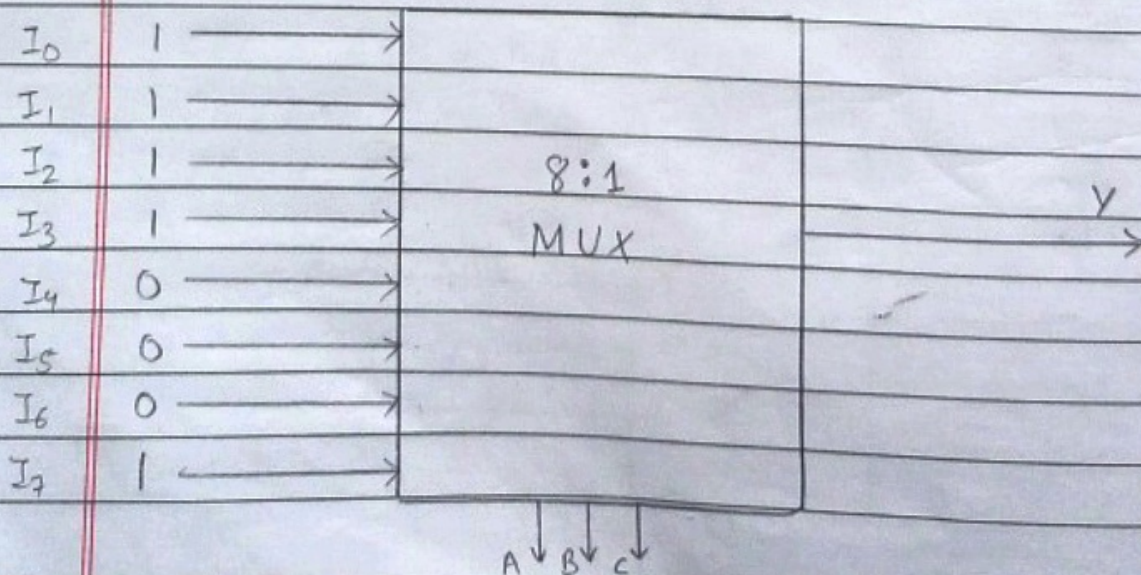
Q3. Implement $Y = \sum m(0, 1, 6, 9, 8, 11, 13)$ using

a) 16:1 MUX

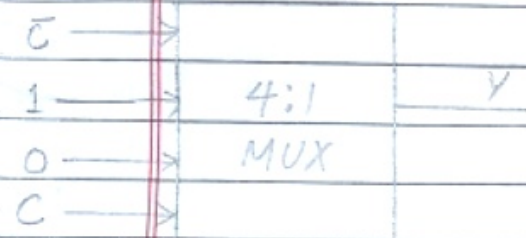
b) 8:1 MUX when B, C, D are selected lines.

①	Y	I	A	B	C	$\bar{A}$	BC	$\bar{A} + BC$
I_0	1		0	0	0	1	0	1
I_1	1		0	0	1	1	0	1
I_2	1		0	1	0	1	0	1
I_3	1		0	1	1	1	1	1
I_4	0		1	0	0	0	0	0
I_5	0		1	0	1	0	0	0
I_6	0		1	1	0	0	0	0
I_7	1		1	1	1	0	1	1

$\sum m(0, 1, 2, 3, 7)$

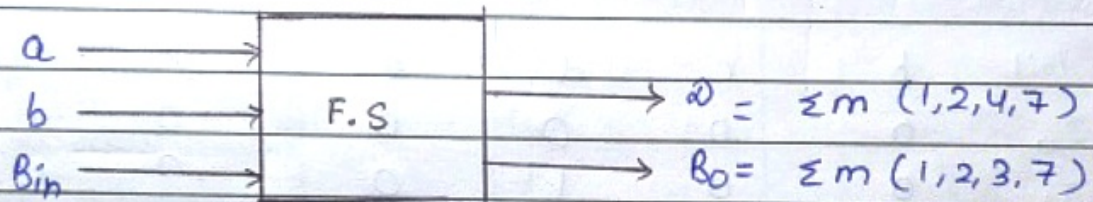


Let A, B be S.L.

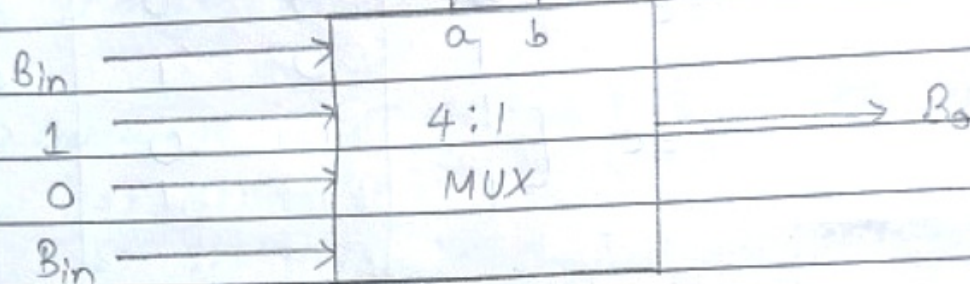
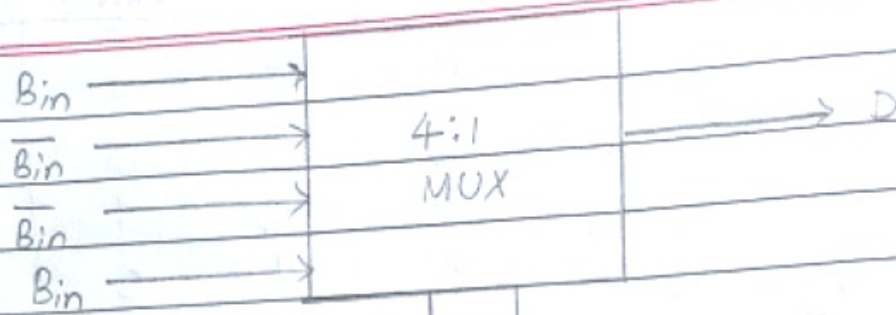


	A	B	C	Y
$Y = \bar{C}$	0	0	0	1
	0	0	1	1
$Y = 1$	0	1	0	1
	0	1	1	1
$Y = 0$	1	0	0	0
	1	0	1	0
$Y = C$	1	1	0	0
	1	1	1	1

②



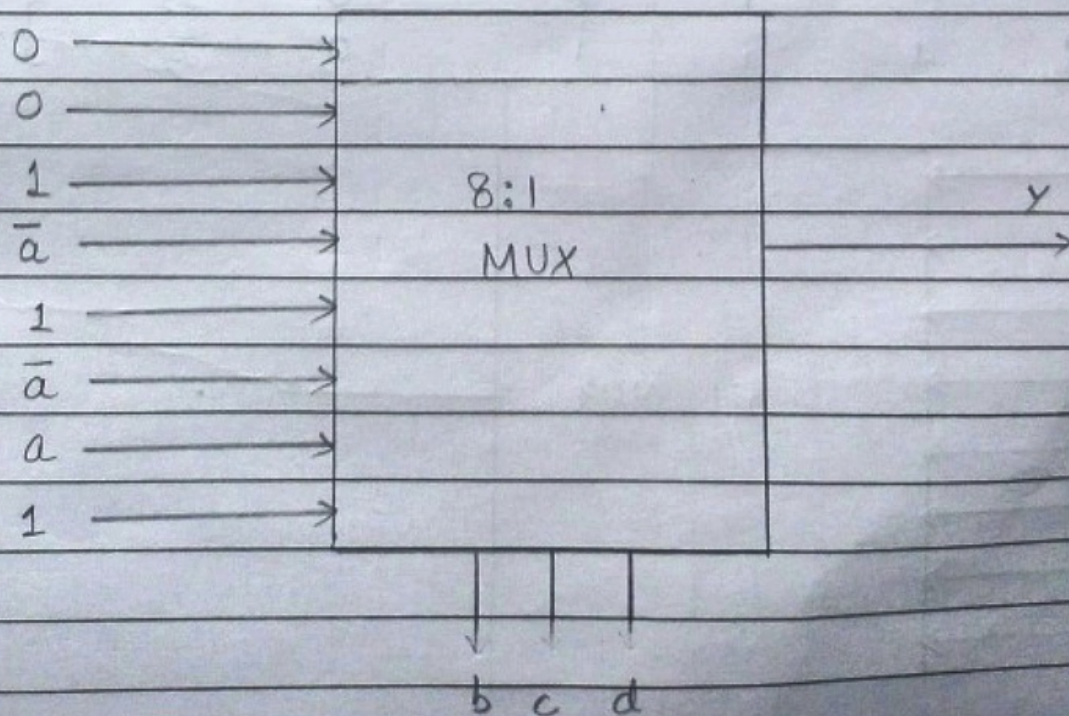
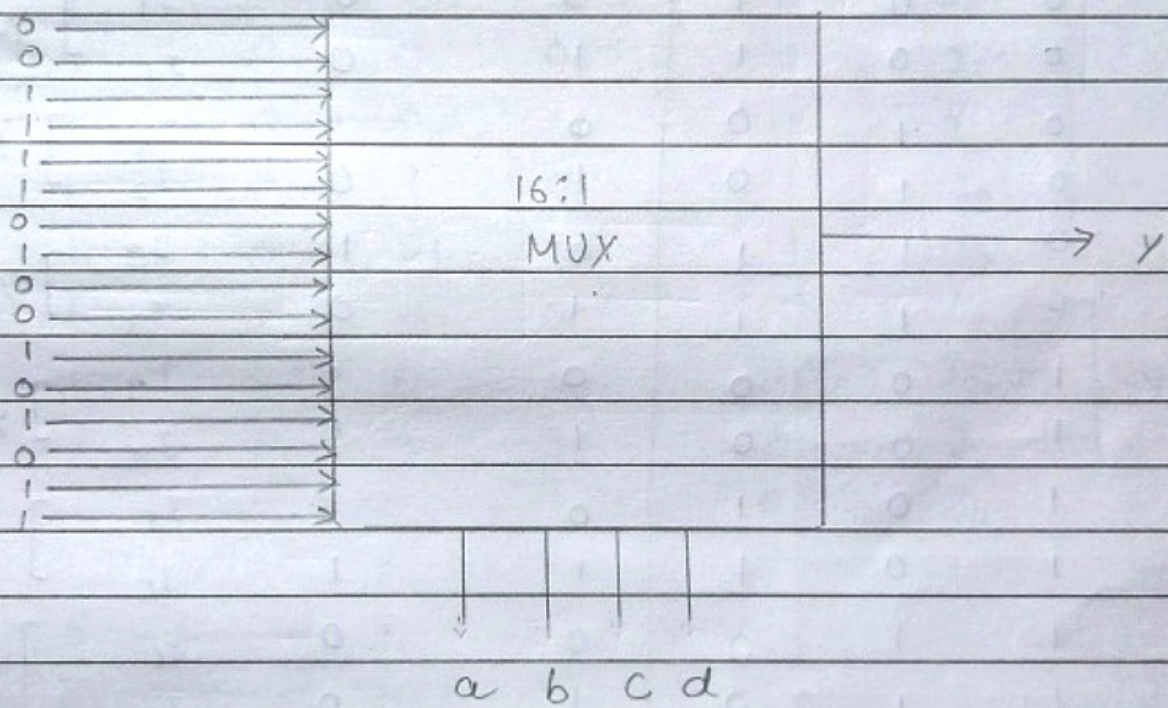
	a	b	Bin	$Y_1 (D)$	$Y_2 (B_0)$
$Y_1 = Bin$	0	0	0	0	0
$Y_2 = Bin$	0	0	1	1	1
$Y_1 = \bar{Bin}$	0	1	0	1	1
$Y_2 = 1$	0	1	1	0	1
$Y_1 = \bar{Bin}$	1	0	0	1	0
$Y_2 = 0$	1	0	1	0	0
$Y_1 = Bin$	1	1	0	0	0
$Y_2 = Bin$	1	1	1	1	1



8:1 (3)

	16:1	b	c	d	a	y
I_0	I_0	0	0	0	0	0
I_1	I_1	0	0	1	0	0
I_2	I_2	0	1	0	0	1
I_3	I_3	0	1	1	0	1
I_4	I_4	1	0	0	0	1
I_5	I_5	1	0	1	0	1
I_6	I_6	1	1	0	0	0
I_7	I_7	1	1	1	0	1
I_8	I_8	0	0	0	1	0
I_9	I_9	0	0	1	1	0
I_{10}	I_{10}	0	1	0	1	1
I_{11}	I_{11}	0	1	1	1	0
I_{12}	I_{12}	1	0	0	1	1
I_{13}	I_{13}	1	0	1	1	0
I_{14}	I_{14}	1	1	0	1	1
I_{15}	I_{15}	1	1	1	1	1

$I_0 \quad Y = 0$
 $I_1 \quad Y = 0$
 $I_2 \quad Y = 1$
 $I_3 \quad Y = \bar{a}$
 $I_4 \quad Y = 1$
 $I_5 \quad Y = \bar{a}$
 $I_6 \quad Y = a$
 $I_7 \quad Y = 1$



Implement following funcⁿ using

a) 8:1 MUX

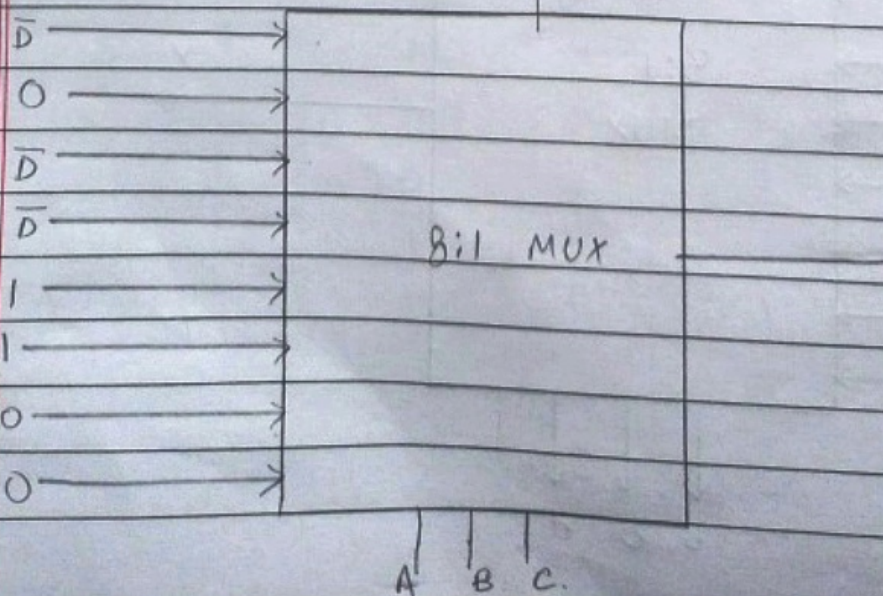
b) 4:1 MUX

(ABC) SL

(AB $\bar{C}$) SL

$$Y = \sum m(1, 2, 3, 5, 7, 12, 13, 14, 15)$$

	4	2	1		
8:1 MUX	A	B	C	D	Y
	0	0	0	0	1 = I_0
	0	0	0	1	0 = I_0
	0	0	1	0	0 = I_1
	0	0	1	1	0 = I_1
	0	1	0	0	1 = I_2
	0	1	0	1	0 = I_2
	0	1	1	0	1 = I_3
	0	1	1	1	0 = I_3
	1	0	0	0	1 = I_4
	1	0	0	1	1 = I_4
	1	0	1	0	1 = I_5
	1	0	1	1	1 = I_5
	1	1	0	0	0 = I_6
	1	1	0	1	0 = I_6
	1	1	1	0	0 = I_7
	1	1	1	1	0 = I_7



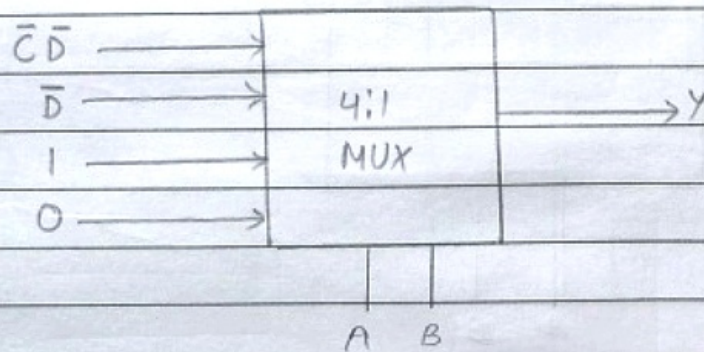
4:1 MUX	A	B	C	D	Y	
	0	0	0	0	1	I_0
	0	0	0	1	0	I_0
	0	0	1	0	0	I_0
	0	0	1	1	0	I_0
	0	1	0	0	1	I_1
	0	1	0	1	0	I_1
	0	1	1	0	1	I_1
	0	1	1	1	0	I_1
	1	0	0	0	1	I_2
	1	0	0	1	1	I_2
	1	0	1	0	1	I_2
	1	0	1	1	1	I_2
	1	1	0	0	0	I_3
	1	1	0	1	0	I_3
	1	1	1	0	0	I_3
	1	1	1	1	0	I_3

$$Y = \overline{C+D} = \overline{C} \overline{D}$$

$$Y = \overline{D} \quad \overline{C} \overline{D} + C \overline{D}$$

$$Y = 1$$

$$Y = 0$$



AB \ CD	00	01	11	10	
00	1				$I_0 = \overline{C} \overline{D}$
01	1			1	$I_1 = \overline{D}$
11					$I_3 = 0$
10	1	1	1	1	$I_2 = 1$

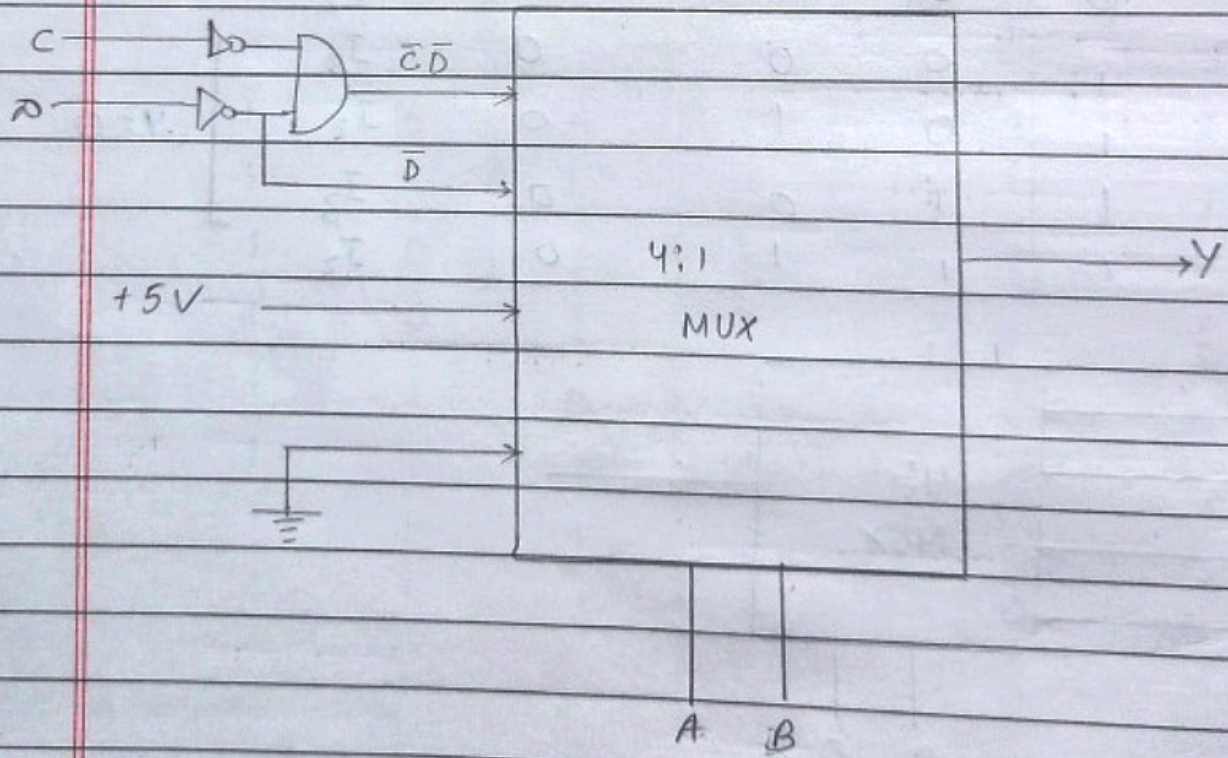
If CD are selected lines, in terms of AB.

AB \ CD	00	01	11	10
00	1			
01	1			1
11				
10	1	1	1	1

I_0 I_1 I_3 I_2
 $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$

$\bar{A} + \bar{B}$ $A\bar{B}$ $A\bar{B}$ $\bar{A}B + A\bar{B} = A \oplus B$

(less maxterm
 Some write in terms
 of maxterm)



*

Multiplexers Tree

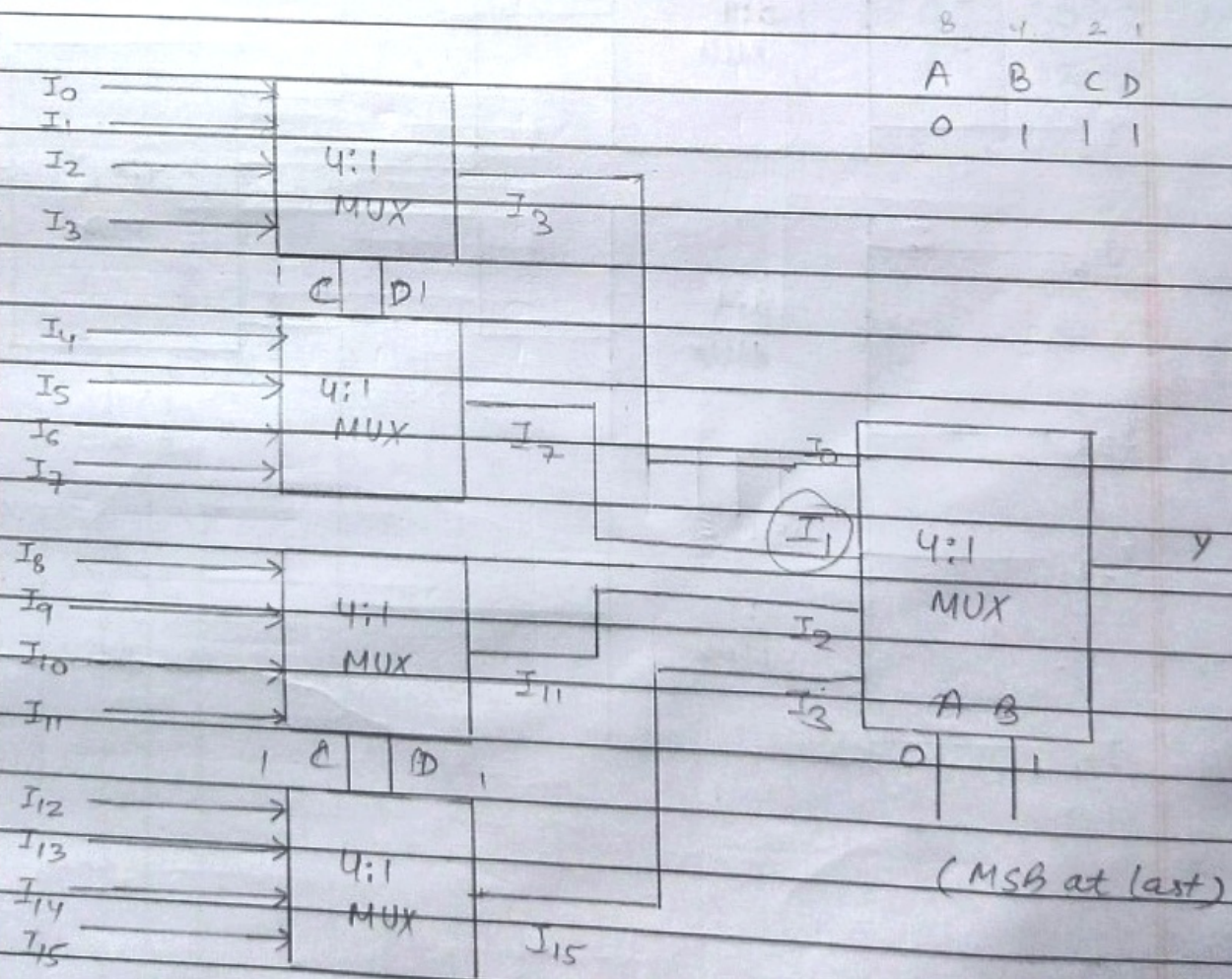
Ques. Implement 16:1 MUX using 4:1 MUX

MUX	I/Ps	S-L	O/Ps
16:1	16	4	1
4:1	4	2	1

Divide till quotient is less than divisor

$$\frac{16}{4} = \frac{4}{4} = 1$$

$$\text{no. of MUX used} = 4 + 1 = 5$$

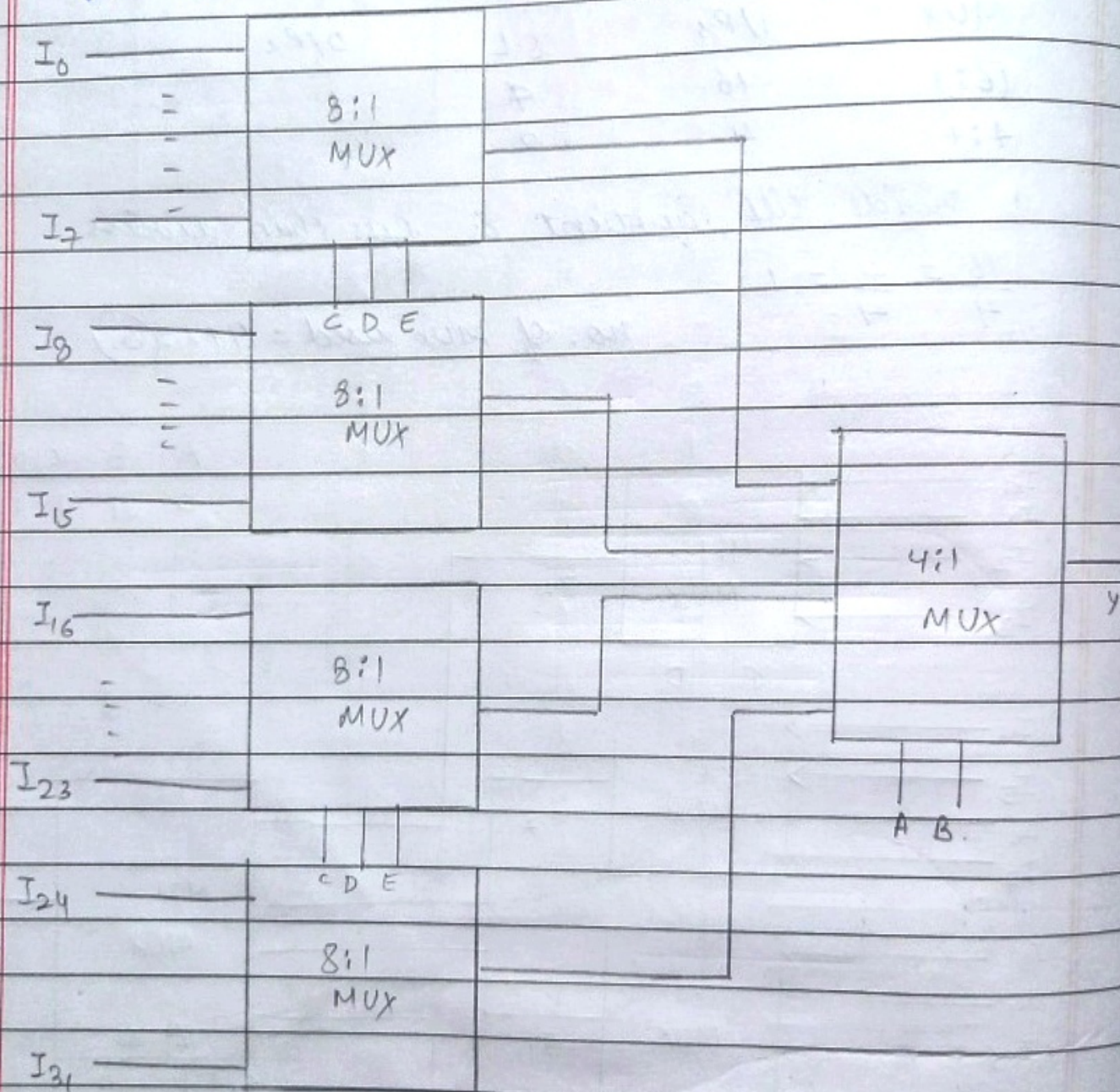


Implement 32:1 MUX using 8:1 MUX and suitable MUX

$$\frac{32}{8} = \frac{4}{4} = 1$$

4 8:1 MUX

1 4:1 MUX



* Demultiplexer

It is the reverse process of multiplexer.

It has one input and many outputs.

It is also called data distributive

$$\text{Total O/Ps} = 2^{SL} \quad (\text{selected lines})$$

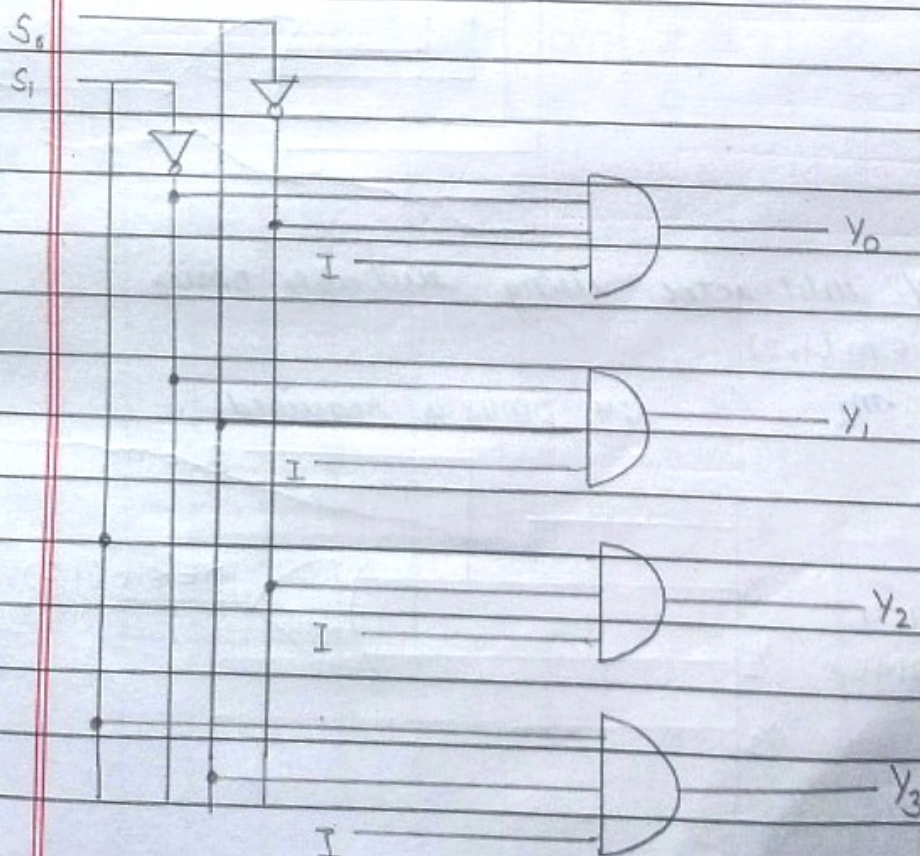
				2	1				
				S_1	S_0	Y_3	Y_2	Y_1	Y_0
I	1,4 DMUX			0	0	0	0	0	I
				0	1	0	0	I	0
				1	0	0	I	0	0
				1	1	I	0	0	0

$$Y_3 = S_1 S_0 I$$

$$Y_2 = S_1 \bar{S}_0 I$$

$$Y_1 = \bar{S}_1 S_0 I$$

$$Y_0 = \bar{S}_1 \bar{S}_0 I$$



To implement any funcⁿ,

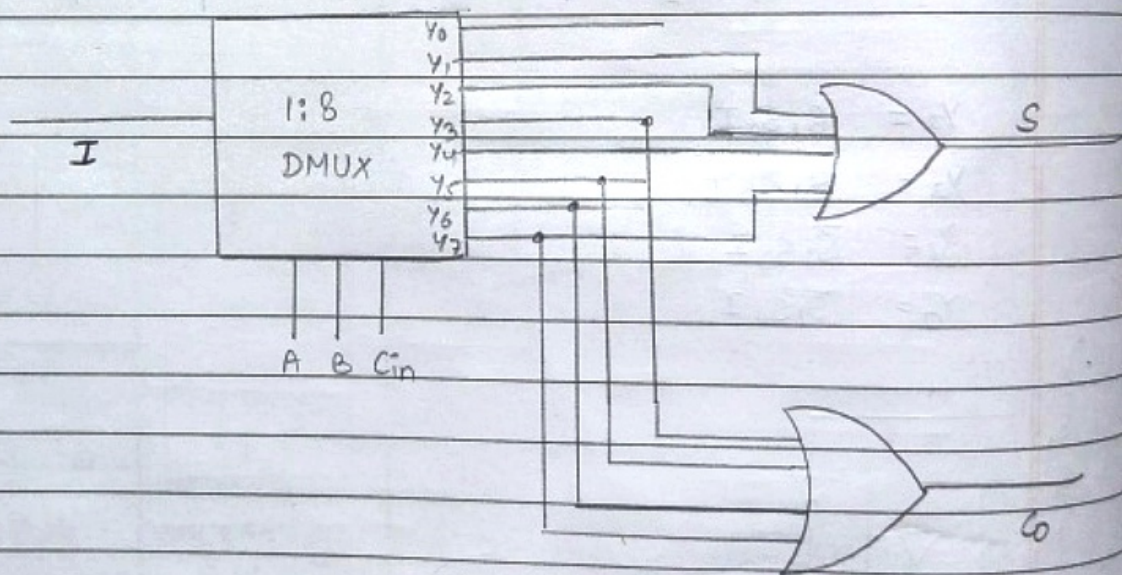
- 1- check max. value of minterm.
- 2- decide required demultiplexer.
- 3- connect all the minterms which are given in problem
- 4- statement to inputs of OR gate.

Ques. Implement full adder using suitable demultiplexer

$$(A, B, C_{in}) \quad S = \sum m(1, 2, 4, 7)$$

$$C_o = \sum m(3, 5, 6, 7)$$

max minterm = 7 so 1:8 DMUX is required.



Ques. Implement half subtractor using suitable DMUX

$$(A, B) \quad D = \sum m(1, 2)$$

$$B_o = m_1$$

1:4 DMUX is required.

